Do Industrial Robots Affect Labor Investment Efficiency?

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Abstract

We examine the impact of industrial robots on the efficiency of firms' labor hiring decisions. Prior research suggests that industrial robots, by providing strong scalability and flexibility, lower a firm's labor adjustment costs. Consistent with this view, our analysis of 1,845 Chinese listed firms from 2010 to 2019 shows that firms with greater exposure to industrial robots make more efficient labor investment decisions. This effect is more pronounced for financially constrained, labor-intensive, and non-state-owned firms. Further analysis reveals that industrial robots enhance labor hiring efficiency particularly in firms with greater production uncertainty and higher labor capabilities. Our findings provide valuable insights for policymakers seeking to maximize the benefits of industrial robot adoption.

Keywords: industrial robot, labor investment efficiency, labor adjustment cost

JEL: E24, J24, J31, J62, O33

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1. Introduction

Industrial robotic automation, driven by fast advancement of artificial intelligence, has substantially reshaped the global labor market (Pratt, 2015). By substituting for both workers and managers, industrial robots transform employment structures and organizational dynamics, with impacts on employment and wages that fundamentally differ from those of traditional capital investments and technologies (Acemoglu and Restrepo, 2020; Dixon et al., 2021; Koch et al., 2021; Duan et al., 2023). Despite these clear implications for workforce dynamics, the consequences of industrial robots for firm labor investment efficiency (LIE) remains underexplored. We fill this gap by investigating how the application of industrial robots affects the efficiency of firms' labor hiring decisions.

Before further discussion, it is important to clarify the distinction between industrial robots (IR) and other forms of automation. According to the International Federation of Robotics (IFR), an industrial robot is "an automatically controlled, reprogrammable, and multipurpose [machine]" (IFR 2014). Acemoglu and Restrepo (2020) further define industrial robots as fully autonomous machines that do not require human operators and can be programmed to perform multiple manual tasks. As such, our study does not include other automation equipment such as conveyor belts, cranes, or machines that are single-purposed or require human operators.

There are two competing views on how industrial robots affect LIE through labor structure changes. The *cost-reduction view* suggests that robots replaces low-skilled workers, increasing demand for high-skilled workers who are typically more costly to adjust (Acemoglu and Restrepo, 2020; De Vries et al., 2020). In contrast, the *quality-improvement view* posits that robots enhance product monitoring and quality control, thereby reducing the demand for middle-skilled workers and managerial staff (Dixon et al., 2021; Bessen et al., 2023). Improvements in operational efficiency under this view may boost demand for low-skilled workers—who typically involve lower adjustment costs—thus enhancing LIE. Distinct from both views, we argue that the impact of industrial robots can go beyond simply substituting for a specific category of workers. Robots can fundamentally change how managers plan production under demand uncertainty and reshape how workers collaborate and contribute to production tasks. They are not merely tools for labor substitution, but strategic enablers of scalability and flexibility which reduce firms' dependency on specific production or management skills.

Industrial robots influence LIE through two key mechanisms. First, robots are scalable: they can be deployed at varying levels of capacity with relatively low marginal cost, allowing firms to adjust production output in response to demand seasonality, uncertainty, and competitive pressure (Mitaritonna and Ragot, 2020; Romanov et al., 2022). Unlike workers, whose hours, contracts, and stamina constrain how quickly their output can expand or contract, industrial robots can easily be added or idled almost instantly, with far fewer frictions such as large-scale hiring or layoffs. Second, robots are flexible: they can be reprogrammed for various tasks and redeployed across business units and locations, allowing firms to reduce bottlenecks, minimize idle time, and avoid overreliance on narrowly specialized labor (Schultz et al., 2003; Sawhney, 2013; Fragapane et al., 2022). Importantly, robot flexibility also complements human capabilities, fostering human-robot collaboration that enhances productivity (Sowa et al., 2021).

Together, scalability and flexibility reduce firms' exposure to labor adjustment costs, including search costs (e.g., job advertising and recruitment), hiring costs (e.g., background checks, interviews, and relocation expenses), training costs, firing costs (e.g., severance pay and legal fees), and productivity disruptions (Yashiv, 2007; Pissarides, 2011; Cao and Rees, 2020). This enables more timely and precise labor hiring decisions. In contrast, firms without robots often struggle to adapt to shifting production needs (Devicienti et al., 2018), and may underreact to market conditions by under-hiring (Azariadis, 1975; Chu and Fang, 2021; Singh, 2022) or under-firing (Lazear, 1990; Autor et al., 2007). Consider, for example, an electronics firm producing goods with varied assembly processes, frequent design updates, and complex tasks like precision soldering and component alignment. Traditionally, such a firm would rely on separate "specialists" for each role and layered managerial oversight, increasing adjustment costs and limiting its ability to scale output when demand changes. With scalable and flexible automation, a robot-enabled firm can operate with a leaner team of core technicians to reprogram and manage robots across multiple lines, supported by generalist workers to complement robot tasks. This shift from specialists to generalists reduces dependence on narrowly specialized workers, hence lowering labor adjustment costs[1] and enhancing the firm's ability to reallocate labor efficiently across tasks and production stages.

[1] Note that this shift does not necessarily affect the labor structure by skill level. In untabulated results, we find that industrial robot usage does not significantly affect the proportion of technicians or educated workers.

We study the firm-specific exposure to robots (FER) of 1845 publicly listed firms in China from 2010 to 2019, and analyze its impact on firm labor investment efficiency. During this period, China records both the highest total sales and the fastest sales growth of industrial robots among the top 8 robots user countries, accounting for almost half of the growth in robots between 2006 and 2016 (Benmelech and Zator, 2022); as such, using the Chinese data for our study is appropriate and representative.[2] We find that firms with greater exposure to robots are associated with significantly more efficient labor investment decisions, as evidenced by less abnormal net hiring. The result holds for both overinvestment and underinvestment measures, alternative measures of labor investment efficiency, and measures of industrial robot exposure estimated using firm annual reports and robot-related patents. In addition, our result remains robust in an instrument variable (IV) regression using the rates of industrial robot penetration in the US and Germany, the two manufacturing powerhouses in North America and Europe. Furthermore, we report that the result is more pronounced for labor-intensive firms and firms with greater financial constraint. Industrial robots appear to primarily help firms avoid under-firing and under-hiring.

To better identify the causality, we consider a quasi-natural experiment involving a government-sponsored pilot program. Participating firms received substantial funding to upgrade their manufacturing execution systems (MES), resulting in increased adoption of industrial robots. Through a difference-in-difference analysis, we find that treatment firms exhibit significantly higher labor investment efficiency post-enrollment compared to non-participating firms. These results underscore a robust and compelling association between the implementation of industrial robots and enhanced labor investment efficiency.

We continue to investigate the underlying mechanisms driving our main findings and find empirical support for both. Consistent with the scalability mechanism, the effects are more pronounced for firms facing heightened competition and sales uncertainty – contexts in which scalable robots provide the greatest value. Consistent with the flexibility mechanism, improvements in LIE are stronger for high-tech firms and firms that hire more educated and skilled

[2] Another motivation to study China stems from the strong incentives to over-hire workers for political considerations in state-owned enterprises (SOEs), according to Du et al. (2012). This further leads to lower labor investment efficiency given the important role of SOEs in China. As is shown in our empirical findings, the adoption of robots in SOEs imposes weaker effects on their labor investment efficiency.

workers, who possess greater adaptability and collaborate more effectively with industrial robots (Chen et al., 2024; Wang et al., 2024).

The paper contributes to the literature by documenting novel findings on how industrial robots affect labor investment efficiency. While prior research primarily focuses on the effects of industrial robots on labor employment and costs, our study highlights their role in optimizing firms' labor hiring decisions. By exploring this aspect, we offer insights into the broader benefits of industrial robots beyond their direct impacts on labor dynamics. Additionally, we provide a comprehensive examination of the mechanisms through which industrial robots influence labor investment efficiency. By showing that scalability and flexibility enable firms to better align workforce adjustments with production needs, our findings assist in identifying which types of firms benefit the most from the adoption of industrial robots. Finally, we show that the improvements in LIE are greater for firms with a larger share of highly capable employees, who demonstrate greater adaptability and collaborate more effectively with robots; this suggests that industrial robots may reshape the way workers collaborate, and this human-robot "synergy" deserves more attention in the literature.

The remainder of the paper is organized as follows. Section 2 reviews the literature and develops testable hypotheses. Section 3 introduces the data and variables. Section 4 discusses the findings. Section 5 concludes.

2. Hypothesis Development

The integration of industrial robots into the manufacturing sector represents a significant shift, transforming work environments and restructuring job activities (Bresnahan et al., 2002; Autor et al., 2003; Brynjolfsson et al., 2019). Industrial robots, particularly when augmented with AI, are increasingly recognized for their critical contributions to enhancing firm productivity and operational efficiencies (Bresnahan and Trajtenberg, 1995; Graetz and Michaels, 2018; Acemoglu and Restrepo, 2020; Koch et al., 2021). Note that firms investing in industrial robots often invest in complementary technologies, so the decision to adopt robots transcends mere capital investment, representing a comprehensive strategic move towards digitalization and automation.

One key factor underpinning this trend is the capability of industrial robots to assume control of repetitive and labor-intensive tasks. This transition to industrial robots enables firms to

reallocate human labor towards more complex and innovative tasks, optimizing the allocation of human resources. Drawing upon this notion, a recent strand of literature discusses the impact of industrial robots on labor decisions. Some specifically focus on the substitutability of AI and robots on work (Autor et al., 2003; Frey and Osborne, 2017; Acemoglu and Restrepo, 2020; De Vries et al., 2020) and predict that industrial robots mainly substitute low-skilled workers, as robots are capable of performing routine, manual task-intensive jobs previously performed by low-skilled workers. This change in employment composition can impact a firm's labor adjustment costs, because employing more high-skilled workers can entail higher labor adjustment costs (Belo et al., 2017; Ghaly et al., 2017; Nguyen and Qiu, 2022); furthermore, higher labor adjustment costs could lead firms to make less efficient labor hiring decisions (Jung et al., 2014; Ben-Nasr and Alshwer, 2016; Cao and Rees, 2020; Khedmati et al., 2020).

By contrast, Dixon et al. (2021) argue that industrial robots are primarily adopted to improve product and service quality instead of cutting costs. Compared to human workers, industrial robots have lower variance in production processes and can substantially reduce a firm's monitoring costs. They show that robots mainly replace middle-skilled workers and managers, due to the decreased need to monitor worker activities to ensure production quality. For low-skilled workers, the adoption of industrial robots could in fact increase the demand and lead to greater employment numbers. Similarly, Bessen et al. (2023) find that firm-level automation mainly affects older and middle-educated workers. If the adoption of industrial robots is mainly affecting skilled workers, it could lead to reduced labor adjustment costs due to the increased proportion of low-skilled workers in the labor force. As such, firms using more industrial robots could make more efficient labor investment decisions.

As discussed before, these findings assume that labor adjustments cost are constant, and the structural changes in labor forces lead to different predictions of labor investment efficiency. However, from a firm's perspective, industrial robots also play an important role in enhancing production scalability and flexibility, allowing firms to quickly adjust production lines to meet changing market demands. Graetz and Michaels (2018) highlight how robots enable manufacturers to swiftly shift between product types without significant downtime, unlike traditional manual setups that require extensive reconfiguration. The ability of robots to operate continuously without the need for breaks or shifts, as discussed by Acemoglu and Restrepo (2020), significantly

enhances production continuity. Autor et al. (2003) argue that robots offer a one-time investment that can replace the recurring costs associated with hiring, training, and managing temporary workers, thereby optimizing labor costs over time. Brynjolfsson et al. (2019) suggest that the robotic workforce mitigates the risk of labor shortages and the associated pressures on wage inflation during peak seasons. The technological advancement brought by industrial robots allows firms to respond dynamically to market demands, maintain continuous production capabilities, and avoid labor adjustment costs associated with frequent hiring and layoffs, training, and production interruptions. These adjustment costs, as well as the uncertainty faced by managers, can lead to underemployment in equilibrium and underreaction to demand shifts (Azariadis, 1975; Chu and Fang, 2021; Singh, 2022). We posit that the flexibility and adaptability brought by industrial robots could also contribute to more efficient labor investment decisions, particularly by enabling firms to better match labor supply with fluctuating demand and reducing the frictions that cause deviations from optimal hiring levels. Therefore, we propose our main hypothesis:

H1: The deployment of industrial robots is positively associated with labor investment efficiency.

Several factors may affect a firm's labor investment efficiency. Cao and Rees (2020) and Khedmati et al. (2020) find that labor intensive firms could be subject to higher labor market frictions. Firms with a heavy reliance on human capital can be more concerns about "talent wars", since they are more likely to face competition for highly skilled workers in sectors that demand intensive human capital. As such, labor intensive firms tend to face greater labor adjustment costs and are less likely to make efficient labor investment decisions (Dolfin, 2006; Chang and Jo, 2019). Similarly, firms under financial constraints often encounter more significant limitations in making labor investment decisions (Benmelech et al., 2021). Additionally, state-owned enterprises (SOEs) typically face government pressure and restrictions regarding labor decisions, while simultaneously possessing fewer incentives to optimize labor investment strategies (Fan et al., 2021) which may hinder their ability to fully leverage technological innovations like industrial robots for optimizing labor strategies (Du et al., 2012; Fan et al., 2021). In contrast, non-state-owned firms often have greater operational flexibility, market-driven incentives, and agility in adopting new technologies, allowing them to realize more substantial efficiency gains from robot deployment. We hence posit that if industrial robots improve labor investment efficiency, the effect

should be more pronounced for labor-intensive firms, financially constrained firms, and SOE firms. Therefore, we propose our secondary hypothesis regarding cross-sectional heterogeneity:

H2: The positive effect of industrial robots on labor investment efficiency is more pronounced for labor-intensive firms, financially constrained firms, and state-owned firms.

Finally, we investigate the two mechanisms previously outlined. Should industrial robots enhance labor investment efficiency by augmenting production scalability, we would expect firms requiring such scalability—namely, those in highly competitive sectors or those experiencing significant sales volatility—would show more pronounced benefits. Furthermore, the flexibility of industrial robots suggests that their benefits are more pronounced when they complement, rather than substitute, for human labor. Prior research shows that employees with greater capabilities, such as high-tech firms and firms that hire more educated and skilled workers, are better equipped to interpret, manage, and adapt to the technological complexity associated with industrial robots (Brynjolfsson and Mcafee, 2017; Acemoglu and Restrepo, 2020). They can leverage robotic systems to enhance coordination, process optimization, and innovation – functions that require cognitive flexibility and problem-solving ability (Autor, 2015). Moreover, human–robot collaboration is most effective when employees possess sufficient technical knowledge and learning capacity to integrate robotic outputs into decision-making and workflow adjustments (Demir et al., 2019; Sowa et al., 2021). Therefore, we propose three hypotheses regarding underlying mechanisms:

H3a (Production scalability): The positive effect of industrial robots on labor investment efficiency is more pronounced for firms with greater product market competition and greater sales volatility.

H3b (Production flexibility): The positive effect of industrial robots on labor investment efficiency is more pronounced for high-tech and firms with a higher percentage of employees with higher education or skill levels.

3. Data and Variables

Our primary data source for industrial robot stocks is the International Federation of Robotics (IFR). Data concerning industrial employment are from the National Bureau of Statistics (NBS). Additional variables were collected from China Securities Market and Accounting Research (CSMAR), a leading provider of financial data in China. Our baseline analysis focuses

exclusively on manufacturing industry A-share public companies. Financial firms and those marked for special treatment (identified with ST, *ST, PT) are excluded from our analysis. Our sample begins in 2010, the earliest year for which industrial robot data is available, and ends in 2019. All continuous variables in our dataset are winsorized at the 1% and 99% levels.

We measure labor investment efficiency (*LIE*) following Jung et al. (2014). We first define labor investment as a firm's net hiring, or the percentage change in the number of employees (Pinnuck and Lillis, 2007). Then we estimate the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment, based on variables such as sales growth, profitability, stock returns, size, leverage, and liquidity. A greater abnormal net hiring value indicates a greater extent of deviation from optimal hiring levels, and hence *less* efficient labor investment decisions. Specifically, the abnormal net hiring, *LIE*, is the absolute value of the residuals from equation (1):

$$\begin{aligned} NetHire_{ijt} = \alpha &+ \sum_{\tau=0}^{1}\beta_{1\tau}SaleGr_{ijt-\tau} + \beta_2 ROA_{it} + \sum_{\tau=0}^{1}\beta_{3\tau}\Delta ROA_{ijt-\tau} \\ &+ \beta_4 Return_{ijt} + \beta_5 SizeRank_{ijt-1} + \beta_6 Lev_{ijt-1} + \beta_7 Quick_{ijt-1} \\ &+ \sum_{\tau=0}^{1}\beta_{8\tau}\Delta Quick_{ijt-\tau} + \sum_{k=0}^{5}\beta_{9\tau}LossBin_{ijt-1}^{k} + \theta_j + \varepsilon_{ijt}. \end{aligned} \tag{1}$$

where subscript *i*, *j* and *t* represent firm, industry, and year, respectively. *SaleGr* is the percentage changes in sales of firm *i* in year *t*. The return on assets, *ROA*, is the net income divided by total assets, and ΔROA is the change in *ROA*. *Return* is the stock's annual return for year *t*. *SizeRank* is the percentile rank of the natural logarithm of the firm's market capitalization. The leverage ratio, *Lev*, is the ratio of long-term debt to total assets. The quick ratio, *Quick*, measures liquidity and is the sum of cash, short-term investment, and receivables divided by current liabilities. $\Delta Quick$ is the change in quick ratio. $LossBin^k$ is a dummy variable which equals 1 if *ROA* falls within the bin from 0 to –0.025, and 0 otherwise. θ_j is the industry fixed effect and ε_{ijt} is the error term.

We measure the adoption of industrial robots with Bartik-type exposure of industrial robots in the spirit of Acemoglu and Restrepo (2020). First, we calculate the rate of industrial robot's penetration (*IRP*) for industry *j* in year *t*, which is given by

$$IRP_{jt} = \frac{IR_{jt}}{L_{jt_0}}. \tag{2}$$

IR_{jt} is the stock of industrial robots of industry j in year t, which is provided by the International Federation of Robotics (IFR). The denominator L_{jt0} is labor employment in industry j in the base year 2010 when robot data becomes available. We next adjust the measure to estimate the firm-level exposure to robots (*FER*) as in equation (3):

$$FER_{ijt} = \frac{WP_{ijt_1}}{\sum_{i \in I_j} WP_{ijt_1}} IRP_{jt} \tag{3}$$

where WP_{ijt} is the number of workers in firm i's production segment in year t. I_j is the set of firms in industry j. The denominator is the aggregate number of workers in manufactory industries of China in year t. In equation (3), the base year t_1 is set at 2011, the earliest year when industry-level data of employment is available. The baseline specification of our model is

$$LIE_{ijt} = \alpha + \gamma FER_{ijt-1} + Ctrl_{ijt-1} + \lambda_i + \delta_t + \nu_{ijt}, \tag{4}$$

where LIE and FER are defined as above. Control variables include the Tobin's q ratio (*Q*), the natural logarithm of the market value (*lnMV*), a dummy indicating whether the firm pays dividends (*DivPaid*), the five-year rolling standard deviation of sales divided by total assets (*sdSales*), the five-year rolling standard deviation of operating cash flows divided by total assets (*sdCFO*), the ratio of property, plant and equipment to the total assets (*Tangible*), a dummy variable indicating whether the firm reports a loss (*Loss*), the proportion of shares held by institutional investors (*Inst*), the five-year rolling standard deviation of net hiring (*sdNetHire*), the number of employees to the total assets (*LabInt*), the non-labor investment inefficiency which is the absolute value of the residual of the regression of investment[3] on sales growth (*abInvest*), the quick ratio (*Quick*), the ratio of long-term debt to total assets (*Lev*), and all first-stage control variables in equation (1). λ_i and δ_t are the firm and year fixed effects.

In this paper, we consider two instrument variables (IV) to address endogeneity issues. These instrument variables are based on the *IRP* in the United States (US) and Germany (DE), which represent the biggest manufacturing powerhouses in North America and Europe, respectively,

[3] Investment is the sum of capital expenditures and R&D expenditure minus the proceeds from the sale of property, plant, and equipment divided by the total assets.

$$IRP_{jt}^{US} = \frac{IR_{jt}^{US}}{L_{jt_1}}, IRP_{jt}^{DE} = \frac{IR_{jt}^{DE}}{L_{jt_1}}.$$

We then define the instrument variables as:

$$IV^{US} = \frac{WP_{ijt_1}}{\sum_{j \in M} \sum_{i \in I_j} WP_{ijt_1}} IRP_{jt}^{US}, IV^{DE} = \frac{WP_{ijt_1}}{\sum_{j \in M} \sum_{i \in I_j} WP_{ijt_1}} IRP_{jt}^{DE}.$$

The US and Germany, in terms of economic size and the completeness of the manufacturing industry chain, are the closest counterparts of China. It is plausible to assume that the application of industrial robots in these countries correlates with that in China; however, it is unlikely that *LIE*s of Chinese public companies will be affected by the use of robots in the US and Germany. We do not use other Asian countries with comparable industrial robot penetration rates, such as Japan and South Korea, because they are adjacent to China and their labor market conditions may resemble that of China. In untabulated results, we obtain similar findings with the *IRP* of the United Kingdom as an additional instrument.

A list of variable definitions is included in Appendix A. We report variable summary statistics in Table 1. Our final sample size is 13,352, and the mean value of *LIE*, our key variable, is 0.168 which is substantial relative to the standard deviation of 0.2683.

[Table 1]

4. Results

4.1. Industrial Robots and Labor Investment Efficiency

We report our benchmark results in Table 2. In columns 1-3, we include basic firm-level control variables, while in columns 4-6 we include additional control variables suggested by Chen et al. (2018) to avoid incorrect statistical inferences. We include firm and year fixed effects in all regressions. In columns 1 and 4, we analyze the full sample, while in columns 2 and 5 (3 and 6) we analyze observations with over-investment (under-investment) in labor only. We see that the coefficient estimates for *FER* are negative and statistically significant at the 5% level in all columns. Since a lower abnormal labor investment value indicates more efficient labor investment decisions, results in Table 2 suggest that the application of industrial robots are associated with more efficient labor investment decisions.

[Table 2]

In Table 3, we perform instrument variable regressions using the two IVs discussed in Section 3. Specifically, we argue that the two instrument variables (*FERs* constructed using the industry robot penetration ratios in the US and Germany) are correlated with the *FER* in China, but should not be strongly correlated with labor investment efficiency in China. The instrument variables we use also passed the weak instrument test and the Hansen's J test, indicating that they are valid instrument in this setting. In all columns, we observe a negative association between *FER* and abnormal net hiring, although in column 2 the statistical significance drops slightly.

[Table 3]

One potential concern is that our findings so far exhibit endogeneity that arise from firm decisions to use robots. If, for example, an omitted variable leads to both more efficient labor investment and greater use of industrial robots, our regression results so far could only show the association but not causality. To address this issue, we consider a quasi-natural experiment based on a pilot program proposed by the Ministry of Industry and Information Technology (MoIIT) of China (Zhu et al., 2024). In 2015, MoIIT first launched the Intellectual Manufacturing Pilot Program, inviting firm applications for government funding support which aims at promoting investment in Intelligent Manufacturing (IM). Firms that enrolled in the pilot program received grants from the local government to upgrade their manufacturing execution system (MES) towards highly automatic production process, which primarily enhances the adoption of industrial robots and supporting systems in such companies. This pilot program was suspended after 2018. In total, 305 companies enrolled in this pilot program from 2015 to 2018 and received funding to purchase/upgrade their industrial robots.

We consider this event as a quasi-natural experiment for industrial robot adoption. We hand-collected the adoption years of this IM pilot program for each company in our sample. To make groups comparable, we first use a propensity score matching model to select control firms. We carefully read the managerial discussion and analysis (MD&A) in firm annual reports to make sure that the control firms do not engage in significant IM upgrading during our sample. We report the covariate imbalance testing of the propensity score matching in Figure 1. In Figure 2, we plot the abnormal net hiring over time for manufacturing firms enrolled in the Intellectual Manufacturing Pilot Program. We see a substantial decline in the abnormal net hiring after firms

enrolled in the pilot program, indicating that the event indeed imposes a significant shock to increase firm labor investment efficiency.

[Figure 1]

[Figure 2]

We then estimate the difference-in-differences (DID) model using this subsample. To deal with the heterogeneous treatment effects, we consider the doubly robust DID estimator proposed by Callaway and Sant'Anna (2021) and the event study estimator proposed by Sun and Abraham (2021). We report the difference-in-difference test results in Table 4. The dependent variable is our *LIE* measure of abnormal net hiring. In columns 1-3 (4-6), we report results using the TWFE (CS) difference-in-difference method. The variable of interest is a dummy variable *IM* which equals 1 if a firm has enrolled in the Intellectual Manufacturing Program, and 0 otherwise. In all columns, the coefficient estimates of *IM* are negative, and statistically significant in 5 out of 6 columns except the overinvestment subsample in column 6. Overall, the difference-in-difference and IV regression results are consistent with the benchmark results. These results support Hypothesis 1 that industrial robots are associated with more efficient labor investment.

[Table 4]

In Appendix C, we report further results on the validity of the DID results. First, we follow Oster (2019) and estimate the Oster's δ, which represents the magnitude of impact of unobserved variables on the treatment effect. A value of δ that is substantially larger than one indicates robust evidence that the treatment effect would persist even if unobserved variables had the same selection effect as the observed variables. We show in Table C1 that the δ values are all greater than 12, indicating that our results are unlikely to be driven by omitted variable bias. Second, in Table C2 we test whether a firm's participation in the IM pilot program is correlated with firm characteristics before and after matching. We show that although a firm's participation in the IM program is associated with several important firm characteristics before matching, the association is substantially reduced after matching, confirming the validity of our methodology. Third, in Table C3 we consider alternative specifications of DID. Following Baker et al. (2022), we use the doubly robust estimator for panel data and report the Stacked DID estimates. We also interact the treatment effect IM with a dummy High LIE, which equals 1 if a firm has higher labor investment

efficiency prior to participating in the IM pilot program, and 0 otherwise. Results in Table C3 are robust and indicate that our results are not driven by the labor investment efficiency levels prior to the treatment effect.

In Table 5, we investigate different types of labor investment (in)efficiency. As Figure 3 shows, inefficient labor investment can be divided into four dimensions: over-hiring, under-firing, over-firing, and under-hiring. The former two are categorized as overinvestment in labor, while the latter two underinvestment in labor. Prior research shows that in the presence of labor adjustment costs and uncertainty, firms may strategically under-react to demand shifts, resulting in under-hiring and under-firing (Azariadis, 1975; Lazear, 1990; Autor et al., 2007; Chu and Fang, 2021; Singh, 2022). If industrial robots provide flexibility to alleviate the hiring inefficiency, we would anticipate the effects to be more pronounced for under-hiring and under firing groups. Results in Table 5 are consistent with our expectation. Specifically, we show that the impact of industrial robots concentrates in under-firing and under-hiring groups; industrial robots help firms make more prompt and effective hiring and firing decisions.

[Table 5]

We next analyze firms that are potentially more vulnerable to inefficient labor investment decisions, which may at the same time benefit more from industrial robots. As discussed in Section 2, we posit that the impact of industrial robots may be more pronounced for labor-intensive, financially constrained, and non-state-owned firms. We report the findings in Table 6. For labor intensity and financial constraints, we divide the sample into subsamples based on whether the characteristic analyzed is above the sample median (the "High" subsample) or not (the "Low" subsample). For state ownership, we investigate subsamples of SOEs and non-SOEs.

[Table 6]

In all columns, the coefficient estimates of *FER* are significantly negative. However, we do see difference magnitude across columns. We use the Chow's F test to test the statistical significance of the differences in coefficients. Our findings suggest that more labor-intensive firms tend to make more efficient labor hiring decisions given the same number of industrial robots deployed, compared to less labor-intensive ones. Similarly, we see more pronounced results for firms that are more financially constrained and are not state-owned, compared to firms that are less

financially constrained and are state-owned. The differences in coefficients across subsamples are statistically significant at the 5% level; our findings are consistent with Hypothesis 2.

4.2. Potential Mechanisms

The association between industrial robots and labor investment efficiency can be attributable to two mechanisms, as discussed previously. In this subsection, we analyze the validity of the two mechanisms. In Table 7, we analyze the first mechanism that is related to the scalability of industrial robots. We posit that, should the mechanism work, the benefits of industrial robots should be stronger for firms that face more volatile sales and fiercer product market competition. This is largely consistent with the coefficient estimates of *FER* in Table 7: while the coefficient estimates are negative in all columns, they are more negative in subsamples of firms with sales volatility and product market competition above sample medians. The F-test suggests that the coefficient differences across subsamples are statistically significant at the 1% level.

[Table 7]

We next consider the flexibility mechanism. As discussed before, flexible industrial robots and employees may complement each other to avoid bottlenecks and develop human-robot collaboration. Because such benefits are more pronounced when employees possess sufficient technical knowledge and learning capabilities so as to adapt to and collaborate with robots, we test if our baseline results are stronger when a firm hires more workers with greater capabilities – namely, high-tech firms and firms that hire more educated and skilled workers. A list of the high-tech industries is included in Appendix A; educated and skilled workers are defined as those with tertiary degrees (some college) and classified as technicians, respectively.

In Table 8, we present subsample analysis by sorting firms into groups by employee capabilities based on high-tech industry status (columns 1 and 2), educated workers (columns 3 and 4), and skilled workers (columns 5 and 6). As shown in Table 8, the results suggest that our results are more pronounced for firms with greater employee capabilities, i.e., with more employees whose education and skill levels allow for more efficient human-labor collaboration. The differences between the high and low subsamples are statistically significant at the 1% level. The findings lend support to H3b and suggest that industrial robots influence labor investment efficiency through the labor flexibility channel.

[Table 8]

4.3. Further Analysis

We further examine the robustness of our results. Specifically, we use two alternative measures of labor investment efficiency for robustness checks. First, we follow Khedmati et al. (2020) and use the abnormal labor cost which is the absolute value of the difference between the actual and expected levels of employee average salary. To estimate the expected employee average salary, we use a model similar to equation (1) and replace the dependent variable with employee average salary. Second, we use a misallocation-based labor investment measure following the seminal work of Hsieh and Klenow (2009); specifically, misallocation is the absolute value of the difference between the actual and economically optimal level of employment, the latter of which is estimated following Hsieh and Klenow (2009). We use these alternative labor investment efficiency measures as our dependent variables in our analysis, and report the results in Table 9. The results from this table are notably consistent with our baseline results, confirming that the efficiency of labor investment is a critical factor in shaping economic outcomes, irrespective of the measure employed.

[Table 9]

For the validity of the independent variable of firm exposure to industrial robots, we consider two robustness measures. The first one is based on the textual analysis of management's discussion and analysis (MD&A) section of annual reports and meanwhile, the second one is the number of robot-related patents. The adoption of industrial robots is an important matter often discussed by managers in their communication with shareholders; hence, we use textual analysis to assess the frequency of robot-related keywords in the MD&A section of firm annual reports, and use this as our first alternative measure of the firm-level adoption of industrial robots. Similar analysis using the MD&A information have been done in the literature to extract key information such as the competition environment (Li et al., 2013; Durnev and Mangen, 2020). Furthermore, we consider the number of patents related to industrial robots as the second alternative measure. Although firms using more robots do not necessarily have more robot-related patents, we argue that firms with more robot-related innovation output (i.e., patents) are also likely to have wider application of industrial robots. We list the “robot-related” keywords used in the MD&A and patent search in Appendix B.

The alternative measures provide a complementary view to our previous analyses, as it captures not only the extent of robot adoption but also the strategic and innovative aspects of firms' engagement with robotics. We report the robustness results in Table 10, which are similar to our baseline results. Overall, the robustness results in Table 9 and Table 10 suggest that our results are not driven by specific measures of labor investment efficiency or firm exposure to industrial robots.

[Table 10]

5. Conclusion

This paper presents novel findings on the impact of industrial robot adoption on labor investment efficiency. Our findings conclusively demonstrate that the integration of industrial robots into the production process significantly enhances the efficiency of labor investment, supporting the hypothesis that technological advancement in the form of robotics is a vital contributor to optimizing labor resource allocation in the manufacturing sector. Notably, the beneficial effects of industrial robots on labor investment efficiency are more pronounced in labor-intensive firms, financially constrained firms, and non-state-owned enterprises. These findings highlight the critical role of industrial robots in providing strategic advantages to firms that are traditionally more vulnerable to inefficiencies in labor management and investment.

This paper also shows that the underlying mechanisms driving these outcomes can be attributed to two main factors: the increased scalability and flexibility afforded by robots. Either of these factors contributes to a more dynamic, responsive, and efficient production environment, where labor resources are optimized, and labor adjustment costs are minimized.

For policymakers, the implications are clear: fostering an environment conducive to technological innovation and adoption in the manufacturing sector is essential. Policies aimed at supporting the integration of industrial robotics can lead to substantial efficiency gains, particularly for firms that face significant labor investment challenges. Furthermore, considering the differentiated impact across firm types, targeted support for labor-intensive, financially constrained, and non-state-owned firms could amplify the positive outcomes of robot adoption. This could include incentives for investment in robotics, support for skills development to complement the technological shift, and measures to encourage research and development in robot-related technologies.

For future researchers, this study opens several avenues for exploration. Investigating the long-term impacts of industrial robot adoption on labor markets, wage dynamics, and even on the broader economy could provide deeper insights into the socio-economic implications of automation. Additionally, exploring the role of human-robot collaboration and examining the transferability of these findings to other sectors beyond manufacturing are critical areas for further research. Understanding the global implications of these dynamics in the context of an increasingly interconnected world economy and in other countries will also be crucial.

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Table 1. Summary Statistics.

This table reports the summary statistics for main variables used in the benchmark regression and instrument variables. The definitions of the variables are referred to Appendix A. All the continuous variables are winsorized at the 1% and 99% levels.

Variable	Obs.	Mean	Std. dev.	Min	Max
Dependent Variables					
LIE	13,352	0.1680	0.2683	0.0015	1.7649
Explanatory Variables & Instrument Variables					
FER CN	13,352	0.3132	0.7001	0.0000	4.6659
FER US	13,352	1.0692	2.0149	0.0000	11.8766
FER GER	13,352	1.3137	2.4370	0.0000	13.9102
Control Variables for Calculating LIE					
saleGr	13,352	0.1612	0.4410	-0.5614	2.8858
ROA	13,352	0.0332	0.0648	-0.2324	0.2185
ΔROA	13,352	-0.0046	0.0615	-0.2610	0.2410
return	13,352	0.1132	0.5065	-0.5957	2.0814
sizerank	13,352	6.0701	2.5724	1	10
lev	13,352	0.0908	0.1071	0.0000	0.5068
quick	13,352	1.2541	1.5102	0.0827	9.6826
$\Delta quick$	13,352	-0.0729	0.8263	-4.5142	2.9948
Control Variables for Explaining LIE					
Q	13,352	2.5279	1.5820	0.8156	9.9863
lnMV	13,352	15.3499	0.9921	13.2175	18.2133
divpaid	13,352	0.6783	0.4672	0	1
sdSales	13,352	0.1341	0.1279	0.0138	0.7808
sdCFO	13,352	0.0531	0.0357	0.0076	0.1971
tangible	13,352	0.2612	0.1571	0.0226	0.7268
loss	13,352	0.1164	0.3207	0	1
inst	13,352	0.4456	0.2372	0.0037	0.9093
sdNetHire	13,352	0.3977	1.0048	0.0066	8.0169
labInt	13,352	0.8154	0.6175	0.0576	3.2908
productivity	13,352	1.1903	1.2762	0.1442	8.4664
abInvest	13,352	0.0421	0.0426	0.0006	0.2361

Table 2. Baseline Results

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of observations with positive abnormal net hiring, while the under-investment sample (labeled "*under*") consists of observations with negative abnormal net hiring. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Sample	full	over	under	full	over	under
FER	-0.0440***	-0.0677**	-0.0259***	-0.0364***	-0.0570**	-0.0205***
	(-5.29)	(-2.47)	(-6.53)	(-4.33)	(-2.17)	(-4.27)
Q	0.0103*	0.0264*	0.0016	0.0032	0.0102	-0.0014
	(1.85)	(1.80)	(0.71)	(1.01)	(1.35)	(-0.99)
lnMV	-0.0217**	-0.0405	-0.0032	0.0131	0.0309	0.0083
	(-2.06)	(-1.54)	(-0.56)	(1.15)	(1.08)	(1.18)
divpaid	-0.0323***	-0.0275	-0.0139**	-0.0172**	-0.0151	-0.0090*
	(-3.54)	(-1.33)	(-2.28)	(-2.13)	(-0.81)	(-1.85)
sdSales	-0.1870***	-0.2589**	-0.0211	-0.1084**	-0.1815	0.0294
	(-3.94)	(-2.24)	(-0.82)	(-2.46)	(-1.59)	(1.57)
sdCFO	0.0651	-0.4430	0.0942	0.0601	-0.3409	0.0409
	(0.46)	(-1.18)	(0.85)	(0.52)	(-1.03)	(0.52)
tangible	-0.0594	-0.0625	-0.0572	-0.0042	0.0482	-0.0182
	(-1.03)	(-0.48)	(-1.51)	(-0.09)	(0.45)	(-0.64)
loss	0.0282**	0.0143	0.0289***	0.0691***	0.0685	0.0675***
	(2.52)	(0.62)	(2.90)	(3.41)	(1.52)	(5.32)
inst	-0.0007	-0.0013	0.0002	-0.0005	-0.0004	-0.0001
	(-1.53)	(-1.24)	(0.76)	(-1.28)	(-0.43)	(-0.46)
sdNetHire	-0.0011	-0.0031	-0.0001	-0.0005	0.0000	-0.0003
	(-1.63)	(-1.20)	(-0.56)	(-0.78)	(-0.01)	(-1.39)
labInt	-0.0590**	-0.1294	0.0398***	-0.0669***	-0.1762***	0.0408***
	(-2.21)	(-1.36)	(3.04)	(-3.32)	(-2.67)	(5.00)
productivity	0.0635	0.2423	0.0171	0.0372	0.2469	-0.0006
	(1.39)	(1.18)	(0.91)	(0.95)	(1.47)	(-0.08)
abInvest	0.0311***	0.0467***	-0.0095***	0.0345***	0.0477***	-0.0051***
	(4.61)	(4.59)	(-3.15)	(4.38)	(4.47)	(-2.82)
quick	0.0022	0.0034	0.0031	-0.0013	-0.0064	0.0045**
	(0.76)	(0.46)	(1.51)	(-0.43)	(-0.91)	(1.97)
lev	-0.0914	-0.1484	0.0693*	-0.1087**	-0.0353	0.0073
	(-1.56)	(-1.01)	(1.82)	(-2.22)	(-0.28)	(0.26)

(*Continued*)	(1)	(2)	(3)	(4)	(5)	(6)
Sample	full	over	under	full	over	under
saleGr				0.2693***	0.3185***	0.2279***
				(27.41)	(15.22)	(20.82)
Δ*ROA*				-0.2025*	-0.3662	-0.3022***
				(-1.82)	(-1.48)	(-4.39)
ROA				-0.0551	-0.0789	0.0627
				(-0.48)	(-0.28)	(0.90)
return				0.0138*	0.0416*	0.0163***
				(1.72)	(1.85)	(3.46)
Δ*quick*				-0.0046	-0.0097	-0.0029
				(-1.27)	(-1.19)	(-1.03)
lagged saleGr				0.0081	0.0162	0.0027
				(0.78)	(0.58)	(0.59)
lagged Δ*ROA*				0.0193	0.0177	-0.0376
				(0.23)	(0.09)	(-0.79)
lagged Δ*quick*				0.0080**	0.0086	0.0032*
				(2.24)	(0.87)	(1.68)
sizerank				-0.0066*	-0.0112	-0.0006
				(-1.82)	(-1.21)	(-0.26)
*LossBin*1				-0.0770***	-0.1266**	-0.0720***
				(-3.12)	(-2.36)	(-3.28)
*LossBin*2				-0.0860**	-0.1376**	-0.0712***
				(-2.56)	(-2.28)	(-3.00)
*LossBin*3				-0.0197	0.0148	-0.0259
				(-0.73)	(0.14)	(-1.42)
*LossBin*4				-0.0511*	-0.0154	-0.0294
				(-1.76)	(-0.20)	(-1.51)
*LossBin*5				0.0163	0.0981	0.0081
				(0.43)	(1.01)	(0.43)
constant	0.5937***	0.9821***	0.1635*	0.0322	-0.1114	-0.0598
	(3.95)	(2.65)	(1.83)	(0.20)	(-0.28)	(-0.60)
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Within R^2	0.0634	0.1187	0.0333	0.2274	0.2595	0.4035
Obs.	13352	4853	8449	13352	4853	8449

Table 3. Instrumental Variables Regressions

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of observations with positive abnormal net hiring, while the under-investment sample (labeled "*under*") consists of observations with negative abnormal net hiring. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The instrument variables are similar to *FER* where the *IRP*s are replaced by those in the United States and Germany. The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
Sample	full	over	under
FER	-0.0524***	-0.0888**	-0.0189**
	(-3.46)	(-1.96)	(-2.11)
All Controls	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Hansen's J	0.1879	0.0994	0.0127
Weak IV test	98.86***	52.22***	78.17***
Obs.	13352	4853	8499

Table 4. Difference-in-Difference Using the Intellectual Manufacturing Pilot Program

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The column labeled "*TWFE-DID*" reports the estimates from two-way fixed effects model. The column labeled "*CS-DID*" reports the doubly robust estimator for panel data proposed by Callaway and Sant'Anna (2021). The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of firms with positive averaged abnormal net hiring before the adoption of intellectual manufacturing, while the under-investment sample (labeled "*under*") consists of firms with negative averaged abnormal net hiring before the adoption of intellectual manufacturing. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. *IM* is a dummy indicating whether the firm participates the intellectual manufacturing pilot program or not. The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	TWFE DID			CS DID		
	full	over	under	full	over	under
IM	-0.1150***	-0.1385**	-0.0820***	-0.1045**	-0.0674	-0.0506***
	(-3.96)	(-2.47)	(-5.22)	(-2.52)	(-1.28)	(-2.78)
All Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	2086	989	1097	1983	874	1089

Table 5. Subsample Analysis: Overinvestment vs. Underinvestment

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The full sample is spilt into four subsamples according to the abnormal and expected hiring. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Overinvestment		Underinvestment	
Sample	Over-hiring	Under-firing	Over-firing	Under-hiring
FER	-0.0466	-0.0483***	-0.0186	-0.0294***
	(-1.35)	(-2.71)	(-1.40)	(-5.39)
All Controls	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Within R^2	0.3500	0.1367	0.1242	0.4498
Obs.	2709	2144	1594	6905

Table 6. Subsample Analysis: Labor Intensity, Financial Constraints, and State Ownership

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The detailed definitions of moderators and control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. Chow's F-test is used to calculate the statistical difference between subsamples. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Moderators	Labor intensive		Financial constraints		State Ownership	
Sample	Low	High	Low	High	SOE	Non SOE
FER	-0.028**	-0.036**	-0.033***	-0.043***	-0.032***	-0.035***
	(-2.49)	(-2.57)	(-3.29)	(-2.83)	(-2.94)	(-2.75)
All Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Differences	0.0079**		0.0106***		0.0035**	
Within R^2	0.2311	0.2685	0.2176	0.2573	0.2214	0.2296
Obs.	6590	6762	6799	6553	5572	7780

Table 7. Subsample Analysis: Sales Volatility and Competition

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The detailed definitions of moderators and control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. Chow's F-test is used to calculate the statistical difference between subsamples. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(1)	(2)
Moderators	Sales Volatility		Product market competition	
Sample	High	Low	High	Low
FER	-0.0603***	-0.0214**	-0.0875***	-0.0261***
	(-3.01)	(-2.40)	(-4.71)	(-2.89)
All Controls	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Differences	-0.0389***		-0.0614***	
Within R^2	0.3014	0.2227	0.2280	0.2451
Obs.	5714	7638	6441	6911

Table 8. Subsample Analysis: Technology, Worker Education, and Skill Levels

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The full sample is spilt into two subsamples according to the moderators. All firm-level control variables in Eq. (4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The detailed definitions of moderators and control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. Chow's F-test is used to calculate the statistical difference between subsamples. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Moderators	High-tech Ind.		%ΔEducated		%ΔSkilled	
Sample	Yes	No	High	Low	High	Low
FER	-0.068***	-0.024***	-0.054***	-0.040***	-0.057***	-0.019**
	(-3.57)	(-2.68)	(-3.72)	(-2.83)	(-3.61)	(-2.17)
All Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Differences	-0.0444***		-0.0139***		-0.0377***	
Within R^2	0.2254	0.2511	0.2368	0.2471	0.2356	0.2669
Obs.	7356	5996	5331	7653	6508	6482

Table 9. Alternative Measures of Labor Investment Efficiency

The dependent variables are the alternative measures of labor investment efficiency (*LIE*) for manufacturing firms: the abnormal labor costs (the absolute value of the difference between the actual and expected level of employee's average salary) in Panel A, and the misallocation of labor forces (the absolute value of the difference between the actual and economically optimal level of employment) in Panel B. The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of observations with positive abnormal net hiring, while the under-investment sample (labeled "*under*") consists of observations with negative abnormal net hiring. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include firm and year fixed effects. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(2). The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. Chow's *F*-test is used to calculate the statistical difference between subsamples. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A. Labor-cost-based LIE

	(1)	(2)	(3)
Sample	full	over	under
FER	-0.0546***	-0.0609***	-0.0501***
	(-4.83)	(-4.54)	(-3.51)
All Controls	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Within R^2	0.0245	0.0432	0.0331
Obs.	13352	4853	8499

Panel B. Misallocation-based LIE

	(1)	(2)	(3)	(4)	(5)	(6)
Est. Method		OP			LP	
Sample	full	over	under	full	over	under
FER	-0.048***	-0.040***	-0.033**	-0.048***	-0.039***	-0.032*
	(-4.18)	(-3.11)	(-2.02)	(-4.03)	(-2.84)	(-1.93)
All Ctrls.	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Within R^2	0.0235	0.2079	0.1962	0.0240	0.2087	0.1980
Obs.	13350	6982	6368	13350	7119	6231

Table 10. Alternative Definitions of Exposure to Industrial Robots

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of observations with positive abnormal net hiring, while the under-investment sample (labeled "*under*") consists of observations with negative abnormal net hiring. *MD&A* is the frequency of robot-related keywords in each firm's MD&A in each year's annual report. *Patent* is the natural logarithm of one plus the number of robot-related patent applications. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Sample	full	over	under	full	over	under
MD&A	-0.057**	-0.075	-0.034*			
	(-2.25)	(-1.08)	(-1.83)			
Patent				-0.063***	-0.099**	-0.020**
				(-3.85)	(-2.02)	(-2.30)
All Ctrls.	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Within R^2	0.2261	0.258	0.4018	0.2265	0.2584	0.4016
Obs.	13352	4853	8499	13352	4853	8499

Figure 1 Covariate Imbalance Testing of Propensity Score Matching

This figure shows the balancing of the covariates between treatment and control groups. Following Anderson et al. (2003) and Chen et al. (2012), we select the covariates relating to cost stickiness. They include natural logarithm of sales (*lnSale*), firm size (*size*), natural logarithm of costs of goods and services (*lnOC*), the proportion of shares held by top 1 shareholder (*top1shd*), return on assets (*roa*), book leverage ratio (*lev*), the proportion of shares held by management teams (*mngtShhld*), the ratio of tangible assets (*tangible*) and the dummies of industry.

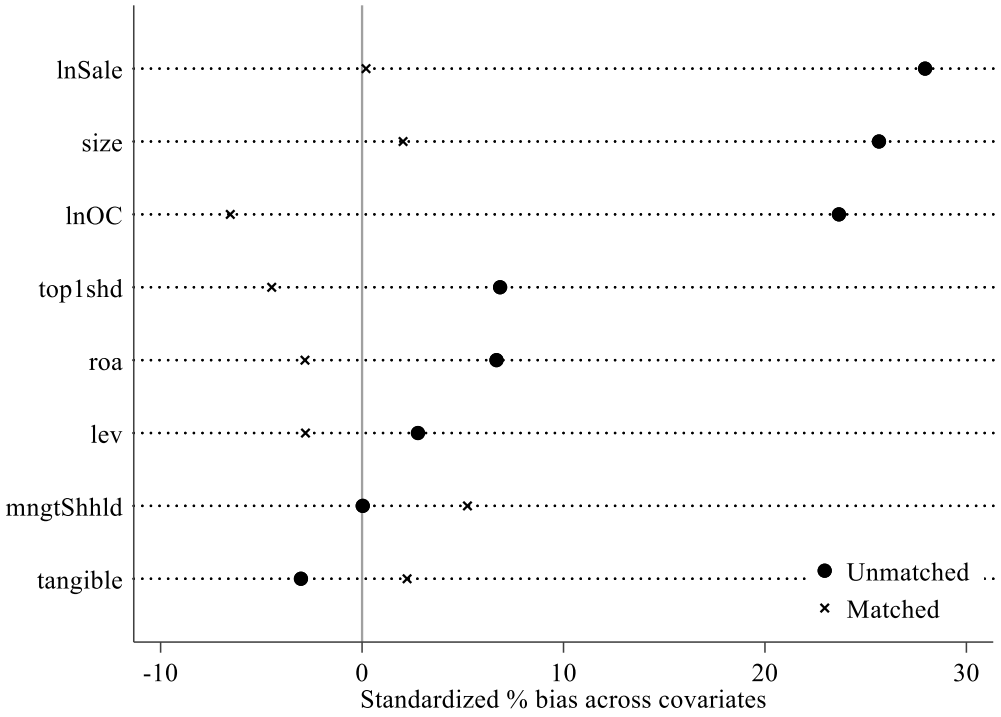


Figure 2 The Intellectual Manufacturing Pilot Program

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The horizontal axis shows the difference between current and adoption year, while the hollow dots represent the coefficients estimated by Sun and Abraham (2021). The spikes attached represent 95% confidential intervals relating with the coefficients.

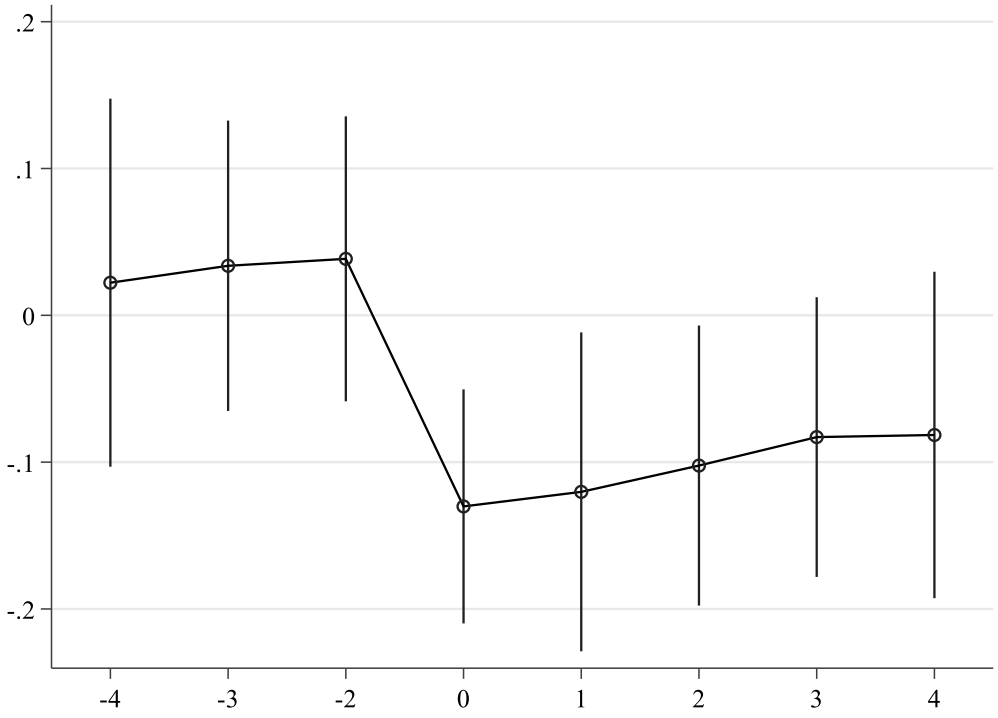
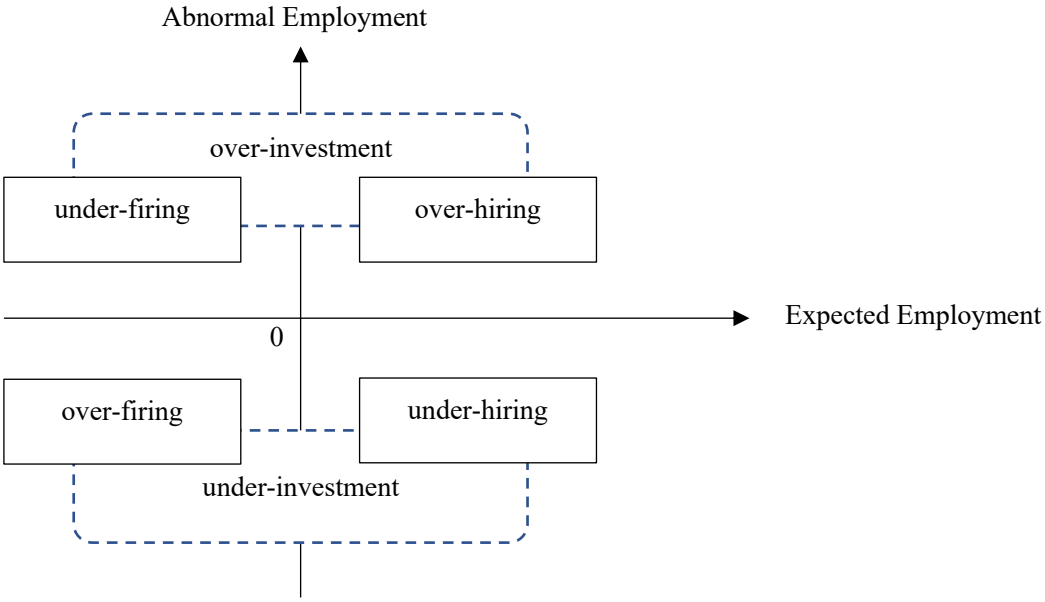


Figure 3 Overinvestment vs. Underinvestment



Appendix A

Category	Variable	Definition
Dependent variables		
	LIE	Labor investment efficiency, the absolute value of the residuals from a regression model of net hiring (the percentage changes of employees) on a set of control variables, which are defined below. A firm is called over- (under-) investment in labor if LIE is larger (smaller) than zero
Explanatory variables		
	IRP	Industrial robot's penetration, the ratio of the stock industrial robot to the labor employment at the beginning of sample period in a given industry
	FER	Firm-level exposure to robot penetration, the weighted average of China's IRP, whereas the weights are the proportion of workers employed in the production departments
Instruments		
	FER US	Firm-level exposure to United States robot penetration, the weighted average of IRP of United States, whereas the weights are the proportion of workers employed in the production departments
	FER DE	Firm-level exposure to German robot penetration, the weighted average of IRP of Germany, whereas the weights are the proportion of workers employed in the production departments
Control variables for baseline specification in equation (4)		
	Q	Market-to-book ratio, the ratio of the market value of equity to the book value of total assets
	lnMV	The natural logarithm of the market value of equity plus the book value of total liabilities
	divpaid	A dummy indicating whether the firm pays dividends in a given year
	sdSales	The five-year rolling standard deviation from t-5 to t-1 of sales-to-assets ratio in a given year t
	sdCFO	The five-year rolling standard deviation from t-5 to t-1 of cash-flows-from-operations-to-assets ratio for a given year t
	tangible	Property, plant, and equipment divided by the total assets
	loss	A dummy indicating whether the firm has RIA
	inst	The proportion of shares held by institutional investors
	sdNetHire	The five-year rolling standard deviation from t-5 to t-1 of employees
	labInt	Labor intensity, the ratio of the number of employees divided by total assets
	productivity	Firm's labor productivity, the ratio of sales to the number of employees
	abInvest	Abnormal investment, the absolute value of the residual of the regression of investment on sales growth, where the investment is defined as the sum of capital expenditures and R&D expenditure minus the proceeds from the sale of property, plant, and equipment divided by the total assets.

(Continued)

Category	Variable	Definition
Control variables for calculating LIE in equation (1)		
	saleGr	Sales growth, the percentage changes in sales
	ROA	Return on assets, the net income divided by total assets
	return	The total annual return for firm's stock
	sizerank	Size rank, a categorical variable of percentile rank of the logarithm of market value of equity
	quick	Quick ratio, the ratio of the sum of cash, short-term investment, and receivables to current liabilities
	lev	Leverage ratio, the ratio of long-term debt to total assets
	LossBink	Loss bin indicator, a dummy variable indicating if the ROA falls between the k-th bin from 0 to tor, a dummy variable indicating i following Pinnuck & Lillis (2007). For example, LossBin1 is equal to 1 if ROA is between [-0.005, 0).
Moderators		
	labor intensity	The ratio of the number of employees divided by total assets. Higher value means the firm is more labor intensive
	financial constraints	The firm-level Whited-Wu Index (WWI). Higher WWI means the firm facing tighter financial constraints
	ownership	A dummy indicating whether the firm is a state-owned enterprise (SOE). A firm is classified as a SOE when it is ultimately controlled by China's central or local governments
	product market competition	The Herfindahl-Hirschman Index (HHI) for all firms in the same industry. Higher HHI means the competition is lower
	sales volatility	The volatility of seasonal component of firm's industry sales, which is measured by the 5-year rolling standard deviation of averaged seasonal components of logarithm sales at the narrowest industry level.
	high-tech industry	The firm-year observation is in the high-tech industry, where the latest (version 2017) high-tech industry classification is extracted from China National Bureau of Statistics. They are C26, C27, C34, C35, C37, C38, C39, C40.
	%ΔEductaed	The percentage change in educated workers hired by the firm. The educated workers are defined as the employees who own a tertiary degree
	%ΔSkilled	The percentage change in technicians hired by the firm. The technicians are defined as the employees who work for the firm's research and technical departments
	excess discretionary expense	The abnormal production costs for as the absolute value of the difference between the actual and expected level of production costs as in Chang et al. (2022).
	accounting quality	The 5-year rolling standard deviation of discretionary accruals from $t-5$ to $t-1$ multiplied by negative one (Jung et al., 2014). The discretionary accruals are the residuals estimated by the modified Jone's model following McNichols (2002).
Quasi-natural experiment		
	IM	A dummy indicating that whether the firm participates China's intellectual manufacturing (*IM*) pilot program or not. It is equal to one once the firm was approved to join in the program.

Appendix B. Industrial Robot Related Keywords

Classification	Chinese Keywords	English Translation
Policy	中国制造 2025	Made in China 2025
	工业 4.0	Industry 4.0
	互联网+	Internet Plus
Features	自动化	Automation
	信息化	Information Oriented
	信息化管理	Information Oriented Management
	信息化应用	Information Oriented Application
	数字化	Digitalization
	网络化	Internet Oriented
	集成化	Integration
	虚拟化	Virtualization
	智能化	Intelligence
Underlying Technologies	物联网	Internet of Things
	虚拟现实	Virtual Reality (VR)
	3D 打印	3D Printing
	人工智能	Artificial Intelligence (AI)
	生物识别	Biological Identification
	模式识别	Pattern Recognition
	神经网络	Neural Networks
	云计算	Cloud Computing
	云平台	Cloud Platform
	云服务	Cloud Services
	云技术	Cloud-related Technology
	大数据	Big Data
	海量数据	Massive Data
	数据中心	Data Center
	数据存储	Data Storage
	数据分析	Data Analysis
	数据挖掘	Data Mining
	互联网	Internet
	移动互联网	Mobile Internet
	互联	Inter-connection
Key Equipment	机器人	Robots
	工业机器人	Industrial Robots
	数控机床	Computer Numerical Control Machine Tool
	数控系统	Computer Numerical Control System
	传感器	Sensors
Commercial Applications	智能物流	Intelligent Logistics
	智能服务	Intelligent Services
	智能终端	Intelligent Terminal
	绿色制造	Green Manufacturing
	高端装备制造	Advanced Equipment Manufacturing
	军民融合	Civil-Military Integration
	智能电网	Smart Energy Grid
	能源互联网	Internet of Energy
		Smart Energy

	智慧能源	Smart Accommodation
	智能家居	Smart City
	智慧城市	Smart Transportation
	智慧交通	Smart Medical Services
	智慧医疗	Smart Community
	智慧社区	E-government
	电子政务	New Energy Vehicles
	新能源汽车	Electric Vehicles
	电动汽车	Power Batteries
	电动车	Charging Points
	动力电池	
	充电桩	

Appendix C. Additional Results

Table C1. Impacts of Potential Omitted Variables.

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of observations with positive abnormal net hiring, while the under-investment sample (labeled "*under*") consists of observations with negative abnormal net hiring. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. The *FER* is the firm-level exposure to industrial robots as the base-year proportion of firm's production workers to the multiplied by its industry robot penetration (*IRP*), which is defined by Eq.(3). The detailed definitions of control variables are reported in Appendix A. The Oster's tests are reported in the last four to two line from the button. The Oster's δ represents that minimal ratio of impact of unobserved variables to that of observed variables such that the coefficient of *FER* excludes zero, under the assumption of the *R*-square from a hypothetical regression of the outcome on both observed and unobserved controls (R^2_{max}) is 1.3 times of estimated *R*-square (R^2), as is suggested by Oster (2019). The Oster's β represents that, under the same assumption, the estimated coefficient of *FER* when the impact of unobserved variables is the same as observed variables. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
Sample	full	over	under
FER	-0.0524***	-0.0888**	-0.0189**
	(-3.46)	(-1.96)	(-2.11)
All Controls	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
$R^2_{max} = 1.3R^2$	0.2934	0.3315	0.5219
Oster's δ	14.0654	19.8972	12.3125
Oster's β	-0.034	-0.0546	-0.019
Obs.	13352	4853	8499

Table C2. Predictivity of Control Variables after and before PSM.

This figure shows the predictivity of the control variables between treatment and control groups after and before PSM. The dependent variable is one if the firm is included in the China's intellectual manufacturing pilot program, and zero otherwise. The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the industry level are used to calculate the *z*-statistics in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
Sample	Post-Match	Post-Match	Pre-Match	Pre-Match
Method	Logit	Probit	Logit	Probit
Q	-0.0121	-0.0080	-0.2300***	-0.4704***
	(-0.25)	(-0.25)	(-5.81)	(-5.11)
lnMV	-0.1024	-0.0680	0.5057***	0.9530***
	(-0.47)	(-0.50)	(6.67)	(6.80)
divpaid	0.1395	0.0947	0.1139	0.2482
	(0.75)	(0.82)	(1.35)	(1.55)
sdSales	-0.1183	-0.1020	0.2368	0.4777
	(-0.10)	(-0.15)	(0.56)	(0.62)
sdCFO	-0.8994	-0.4793	-1.6755	-3.4144
	(-0.24)	(-0.22)	(-1.44)	(-1.64)
tangible	-0.5189	-0.3348	-0.0026	0.0697
	(-0.44)	(-0.46)	(-0.01)	(0.09)
loss	-0.2594	-0.1633	-0.0681	-0.1323
	(-1.37)	(-1.40)	(-0.82)	(-0.86)
inst	0.0112	0.0067	0.0052**	0.0096**
	(1.45)	(1.46)	(2.52)	(2.32)
sdNetHire	-0.0065	-0.004	-0.0086	-0.0181
	(-0.94)	(-0.97)	(-0.67)	(-0.66)
labInt	0.5031	0.2878	0.0371	0.0548
	(1.35)	(1.35)	(0.55)	(0.42)
productivity	0.0592	-0.0536	0.2971***	0.5852***
	(0.04)	(-0.06)	(2.59)	(2.58)
abInvest	0.028	0.0153	-0.0278*	-0.0503**
	(0.38)	(0.38)	(-1.85)	(-2.00)
quick	-0.1911***	-0.1098***	-0.0559	-0.1225
	(-2.79)	(-3.13)	(-1.43)	(-1.33)
lev	-0.4309	-0.2609	0.0384	0.0544
	(-0.22)	(-0.22)	(0.08)	(0.05)
constant	-0.1435	0.0154	-9.2071***	-17.1669***
	(-0.04)	(0.01)	(-7.69)	(-7.89)
Industry dummy	Yes	Yes	Yes	Yes
Pseudo R^2	0.0685	0.0679	0.2006	0.1971
Obs.	2394	2394	12537	12537

Table C3. Alternative Specifications of DID

The dependent variables are the labor investment efficiency (*LIE*) for manufacturing firms, which is the abnormal net hiring as the absolute value of the difference between the actual and expected level of labor investment in Eq.(1). The column labeled "*DDD*" reports the estimates from the difference-in-differences-in-differences model. The column labeled "*Stacked DID*" reports the doubly robust estimator for panel data suggested by Baker et al. (2022). The full sample (labeled "*full*") contains all non-missing firm-year observations. The over-investment sample (labeled "*over*") consists of firms with positive averaged abnormal net hiring before the adoption of intellectual manufacturing, while the under-investment sample (labeled "*under*") consists of firms with negative averaged abnormal net hiring before the adoption of intellectual manufacturing. All firm-level control variables in Eq.(4) are lagged by one year. All specifications include untabulated firm and year dummies. *IM* is a dummy indicating whether the firm participates the intellectual manufacturing pilot program or not. *High LIE* is one if the firm's 3-year averaged *LIE* before the inclusion of pivot program is lower than the sample median. The detailed definitions of control variables are reported in Appendix A. Standard errors clustered at the firm level are used to calculate the *t*-value in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	DDD	Stacked DID		
		full	over	under
IM	0.0429	-0.1064***	-0.0805	-0.0670***
	(1.60)	(-4.08)	(-1.33)	(-3.59)
High LIE	0.1708***			
	(8.61)			
IM×High LIE	-0.1427**			
	(-2.26)			
All Controls	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Within R^2	0.1685	0.1017	0.1835	0.0595
Obs.	2086	12343	4606	7737